Bihar Engineering University, Patna

B.Tech 1st Semester Examination, 2025

Course: B.Tech
Code: 100110

Subject: Engineering Physics

Time: 03 Hours
Full Marks: 70

Instructions: -

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Displacement vs. velocity graph in case of simple harmonic motion is
(i) Straight
(ii) Circular
(iii) Hyperbolic
(iv) Elliptical

(b) Paramagnetic materials are
(i) Weakly repelled by magnetic field
(ii) Strongly repelled by magnetic field
(iii) Weakly attracted by magnetic field
(iv) Strongly attracted by magnetic field

(c) Which phenomenon explains the wave nature of light?
(i) Photoelectric effect
(ii) Pair production
(iii) Blackbody radiation
(iv) None of the above

(d) Which of the following atomic state has longest life time?
(i) Excited State
(ii) Metastable state
(iii) Intermediate state
(iv) Ground state

(e) Write down four medical applications of LASER.

(f) Young's double slit experiment was performed in a laboratory by taking monochromatic blue, orange and red light and fringe widths were obtained as β_B, β_O, β_R respectively. Other variables had been kept constants. Write down the fringe widths in increasing order.

(g) Give two examples of Non-Ohmic conductors.

(h) The de Broglie wavelength of a particle moving with a velocity 200 m/s is $14.14A^0$. What will be de Broglie wavelength if kinetic energy is doubled?

(i) State Heisenberg's uncertainty principle.

(j) What is the ratio of intensities of maxima and minima in interference if the intensity ratio is 25 : 1 ?

Q.2 (a) Derive the differential equation for Damped Harmonic Oscillator. Give its solution. Differentiate between under damped, over damped and critically damped harmonic oscillation including graphical representation. **[10]**

(b) The time period of simple harmonic oscillator is 2s. It is subjected to a damping force proportional to speed with damping co-efficient $\beta = 1 sec^{-1}$. Find the time period and logarithmic decrement. What should be the value of β for the motion to be critically damped? **[4]**

Q.3 (a) Discuss the structure and working of Ruby LASER. Draw and explain its energy level diagram. **[10]**

(b) Write notes on pumping mechanisms in LASER. **[4]**

Q.4 (a) What is diffraction grating? Discuss the diffraction pattern due to a grating with intensities distribution curve. [10]

(b) Write a short note on Michelson's interferometer. [4]

Q.5 (a) Discuss Kronig-Penney model for the motion of an electron in a periodic potential and the existence of allowed and forbidden energy bands. [10]

(b) Discuss the classification of solids into metals, insulators and semiconductors on the basis of energy band theory. [4]

Q.6 (a) State Biot-Savart's law. Apply it to derive an expression for magnetic flux density at a point near an infinitely long straight current carrying conductor. [10]

(b) Derive an expression for torque on a magnetic dipole placed inside a uniform magnetic field. [4]

Q.7 Write down Schrödinger's wave equation for a particle inside one dimensional potential well given by, [14]

$V(x) = 0, \quad 0 < x < a$

$= \infty, \quad x < 0$ and $x > a$

Solve it to obtain energy eigen values and eigen functions.

Q.8 (a) Define Poynting vector. Use Maxwell's equations to derive Poynting theorem. [10]

(b) State Ampere's circuital law. Write down the corresponding Maxwell's equation. [4]

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]

(a) Coriolis effect

(b) Electronic polarization in dielectrics

(c) Compton scattering